

NATIONAL JUNIOR COLLEGE
PRELIMINARY EXAMINATION
Higher 1

CANDIDATE
NAME

SUBJECT
CLASS

REGISTRATION
NUMBER

CHEMISTRY

Paper 2 Structured Questions and
Free Response Questions

Additional Materials: Writing Paper
Data Booklet

8872/02

Friday 14 Sep 2012
2 hours

READ THE INSTRUCTION FIRST

Write your subject class, registration number and name on all the work you hand in.

Do not open this booklet until you are told to do so.

Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working. Do not use paper clips, highlighters, glue or correction fluid.

Section A:

Answer **ALL** questions in the spaces provided on the question paper.

Section B:

Answer **two** questions on separate answer paper.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, fasten all your work together.

You are advised to spend one hour on each section of the paper.

For Examiner's Use		
Section A	1	/12
	2	/10
	3	/10
	4	/8
Section B	5	/20
	6	/20
	7	/20
Total		/80

Section A: Structured Questions [Total: 40 marks]

Answer **ALL** questions in the space provided.

- 1** An unknown hydrocarbon **X**, can be obtained from the catalytic cracking of long chain hydrocarbons from the refining of crude oil.

Compound **X** has the following composition by mass: C: 85.7%, H 14.3%.

Pure **X** is usually stored as a liquid under pressure. When 0.126 g sample of pure **X** was vapourised in a suitable apparatus, the vapour occupied 55.0 cm³ at 25°C and 101 kPa. The following ideal gas equation can be used.

$$pV = \frac{m}{M_r} RT$$

where p = pressure in Pa, V = volume in m³, m = mass in grams, M_r = molecular mass of compound, R = ideal gas constant of 8.31 JK⁻¹mol⁻¹ and T = temperature in Kelvins.

A, **B** and **C** are three possible structures of compound **X**. It was found that all 3 compounds decolourize aqueous bromine. When acidified KMnO₄ was added into the 3 compounds and heated, only **A** produces a colourless gas and a compound **D** which does not form an orange precipitate with 2,4-dinitrophenylhydrazine while **B** and **C** produces ethanoic acid as the only product.

The melting points and boiling points of **B** and **C** are as shown below:

	Boiling Point / °C	Melting Point / °C
B	4	-139
C	1	-106

- (a) (i) Calculate the empirical formula of compound **X**.

[1]

- (ii) Using the ideal gas equation, find the M_r of compound **X** and hence deduce its molecular formula.

[3]

- (b) (i) State the functional group present in **A**, **B** and **C** based on the reactions above.

.....
.....

[1]

- (ii) From the information given above, deduce the structures of **A**, **B** and **C**.

[3]

- (iii) Identify the gas evolved and the compound **D** that is produced when acidified KMnO_4 was added into **A**. Hence write a balanced equation for it.

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[2]

- (iv) Explain why **B** has a higher boiling point but lower melting point than **C**?

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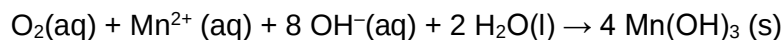
[2]

[Total:12]

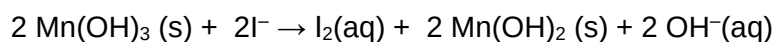
- 2 The Dissolved Oxygen Concentration (DOC) in rivers and lakes is important for aquatic life. If DOC falls below 5 mg dm^{-3} , most species of fish cannot survive.

Environmental chemists can determine the DOC in water using the procedure below.

A sample of water is shaken with aqueous Mn^{2+} and aqueous alkali. The dissolved oxygen oxidises the Mn^{2+} to Mn^{3+} , forming a pale brown precipitate of $\text{Mn}(\text{OH})_3$.



The $\text{Mn}(\text{OH})_3$ precipitate is then reacted with an excess of aqueous potassium iodide, which is oxidized to iodine, I_2 .



The iodine formed is then determined by titration using aqueous sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$.

- (a) A student decided to analyse the National Junior College's pond water using the above procedure. The student found that a 20 cm^3 sample of the pond water required 18.6 cm^3 of $0.00100 \text{ mol dm}^{-3}$ of $\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$ for titration.

- (i) Calculate the DOC of the sample of river water, in mg dm^{-3} .

- (ii) Comment on whether there is enough dissolved oxygen in the pond water for the carps to survive.

.....

.....

[4]

- (b) The presence of nitrite ions, NO_2^- , interferes with this method because NO_2^- ions can also oxidize iodide ions into iodine. During the reaction, a colourless gas, **Y**, with a molar mass of 30 g mol^{-1} is also produced.

(i) Predict the formula of the colourless gas.

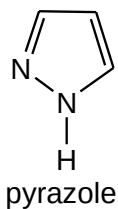
(ii) Draw the dot-and-cross diagrams for the colourless gas, **Y**, and NO_2^- . Suggest the shape of each of these molecules.

(iii) Write an equation for the oxidation of aqueous iodide ions by aqueous nitrite ion. Hydroxide ions are produced in this reaction.

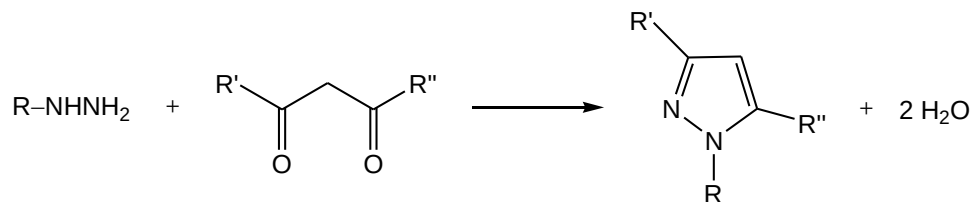
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[6]
[Total:10]

3 Many pharmaceutical molecules contain the pyrazole ring system.



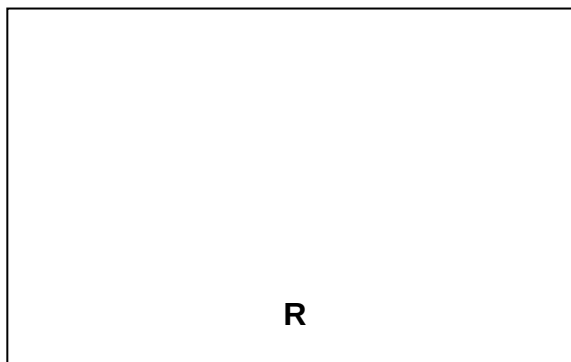
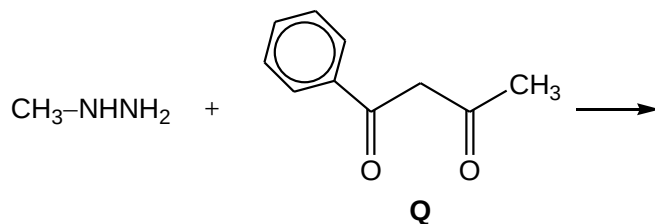
(a) Substituted pyrazoles are often synthesized by reacting a hydrazine and a diketone:



(i) State the type of reaction between the hydrazine and diketone.

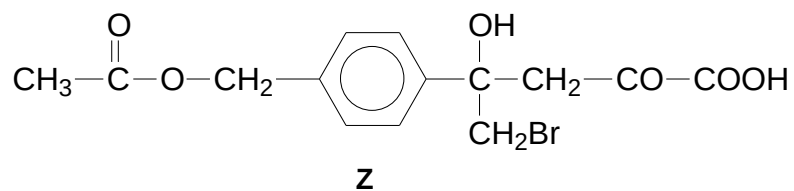
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(ii) Draw the structure of the product **R** formed when CH_3NHNH_2 reacts with diketone **Q** in a similar reaction.



[3]

(b)

(i) Name the four functional groups in compound **Z**.

1)

2)

3)

4)

(ii) Draw the structural formulae of the compound(s) when compound **Z** is treated with

○ Cold NaOH(aq)

○ Hot NaOH (aq)

- LiAlH_4 in dry ether

[7]
[Total:10]

- 4 The melting points of some of the oxides formed by Period 3 elements are given in a random order below.

Oxide	A	B	C	D	E
Melting point/ °C	2852	-73	1610	1275	300

- (a) (i) Using the letters **A** to **E**, give two oxides which have simple molecular structures.

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- (ii) Give a simple chemical test that could be used to show which of the oxides in the table sodium oxide is. State the observation you would make.

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[3]

- (b) (i) Write the full electronic configuration for the sodium ion, Na^+ .

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- (ii) Explain why the second ionisation energy of sodium is greater than that of magnesium.

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- (iii) By quoting appropriate data from the *Data Booklet*, explain why sodium oxide has a lower melting point than magnesium oxide.

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[5]
[Total:8]

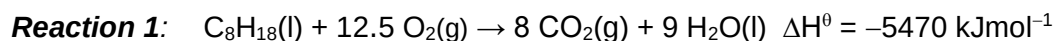
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Section B: Free Response Questions [Total: 40 marks]

Answer **two** of three questions in this section on separate paper.

- 5** Gasoline or petrol is a transparent petroleum-derived liquid that is primarily used as a fuel in internal combustion engines. It consists mostly of organic compounds obtained by the fractional distillation of petroleum, enhanced with a variety of additives.

- (a) Energy is obtained from the combustion of gasoline by the conversion of a hydrocarbon to carbon dioxide and water. The combustion of octane follows this reaction:



- (i) Use the value of ΔH° given above and the data below to calculate the standard enthalpy change of formation of octane.

Compound	$\Delta H^\circ_f / \text{kJ mol}^{-1}$
CO_2	-393.5
H_2O	-285.8

- (ii) Suggest the type of hybridisation of the carbon atoms in octane and carbon dioxide. [3]
- (b) (i) Use suitable bond energy values from the *Data Booklet* to calculate the standard enthalpy change of **Reaction 1**.
- (ii) Suggest two reasons for the difference between your calculated value and that given in **Reaction 1**. [4]
- (c) The carbon dioxide produced dissolves in water to form a monobasic acid, carbonic acid, H_2CO_3 , which has a pH of 3.5.
- (i) Calculate the concentration, in mol dm^{-3} , of hydrogen ions in carbonic acid.
- (ii) 25.0 cm^3 of a sample of the carbonic acid was titrated with 0.25 mol dm^{-3} aqueous NaOH. 21.25 cm^3 of the aqueous NaOH was required to reach equivalence point.

Calculate the concentration, in mol dm^{-3} , of the carbonic acid.

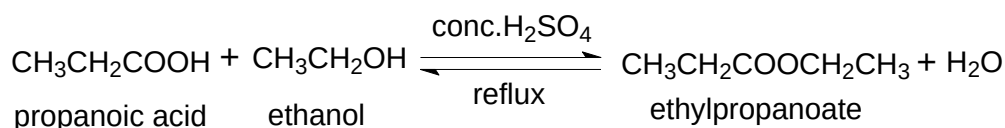
- (iii) Based on your answers to (c)(i) and (c)(ii) above, what can you deduce about the strength of carbonic acid? Give a reason for your deduction.
- (iv) Suggest a suitable indicator for the above titration. [5]

- (d) Carbonic acid and its salt have an important role in controlling pH in blood.
- (i) Explain what is meant by a *buffer solution*.
 - (ii) With the aid of two equations, explain how H_2CO_3 and its salt can control pH in the blood. [3]
- (e) An alkane **F** is a component of petrol. When treated with bromine under free radical conditions it produces a mixture of brominated compounds, including **G**, $\text{C}_6\text{H}_{12}\text{Br}_2$. Dehydrobromination of **G** produces **H**, C_6H_{10} . When **H** is oxidised by hot concentrated KMnO_4 , three compounds, CO_2 , $\text{CH}_3\text{CO}_2\text{H}$ and $\text{CH}_3\text{COCO}_2\text{H}$, are formed in equimolar amounts.
- (i) Suggest reagents and conditions for the conversion of **G** to **H**.
 - (ii) Suggest a structure for compound **H**, and hence deduce the structure of compound **F**.
 - (iii) There are four possible isomers of $\text{C}_6\text{H}_{12}\text{Br}_2$ that could give **H** on dehydrobromination. Draw the structures of **two** of them. [5]

[Total:20]

6 Carbon, hydrogen and oxygen are three common elements found in organic compounds

- (a) One class of organic compounds containing carbon, hydrogen and oxygen are esters. An example of an ester, ethyl propanoate, $\text{CH}_3\text{CH}_2\text{COOCH}_2\text{CH}_3$, is generally used for replicating the scents or flavors of fruits such as blackberry and pineapple. It is made by heating propanoic acid and ethanol in the presence of concentrated sulfuric acid.



5.8 g of $\text{CH}_3\text{CH}_2\text{COOH}$ is mixed with 3.0 g of $\text{CH}_3\text{CH}_2\text{OH}$ and heated under reflux in the presence of concentrated H_2SO_4 until the above equilibrium is established.

The amount of $\text{CH}_3\text{CH}_2\text{OH}$ at equilibrium is found by titrating it against acidified 0.5 mol dm^{-3} $\text{K}_2\text{Cr}_2\text{O}_7$. It was found that 40.00 cm^3 of $\text{K}_2\text{Cr}_2\text{O}_7$ was required.

- (i) Write a balanced equation for the reaction between $\text{CH}_3\text{CH}_2\text{OH}$ and acidified $\text{K}_2\text{Cr}_2\text{O}_7$.
- (ii) Write a K_c expression for the above reaction and hence determine its value.
- (iii) Another student attempted to find the number of moles of $\text{CH}_3\text{CH}_2\text{OH}$ remaining at equilibrium by titrating the solution against 0.5 mol dm^{-3} acidified KMnO_4 instead.

Find the volume of KMnO_4 required using your answer from (ii).

- (iv) If the titration in (iii) is carried out using alkaline KMnO_4 , state and explain if the volume of KMnO_4 used would still be the same. [10]

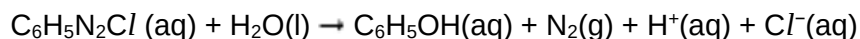
- (b) Compound **A** has molecular formula, $\text{C}_6\text{H}_{10}\text{O}_2$. Reaction of **A** with hot dilute hydrochloric acid produces compound **B**, $\text{C}_4\text{H}_8\text{O}$ and **C**, $\text{C}_2\text{H}_4\text{O}_2$. Reaction of **B** with hot acidified potassium manganate(VII) produces compound **D**, $\text{C}_3\text{H}_4\text{O}_4$, and effervescence is observed. One mole of **D** requires two moles of NaOH for complete neutralisation. **B** decolourises aqueous bromine rapidly to form compound **E**, $\text{C}_4\text{H}_9\text{O}_2\text{Br}$ as the major product. Both **B** and **E** turn hot orange acidified potassium dichromate(VI) green. When treated with phosphorus pentachloride, **B** gives white fumes and compound **F**, $\text{C}_4\text{H}_7\text{Cl}$, is formed.

Suggest a possible structure for each of the compounds **A** to **F**. Explain the chemistry of the reactions described.

[10]
[Total:20]

- 7 Benzenediazonium chloride is an organic compound with the formula $\text{C}_6\text{H}_5\text{N}_2\text{Cl}$. It exists as a colourless crystalline solid that is soluble in polar solvents. It is representative of a family of aryldiazonium compounds that are used to prepare azo dyes.

- (a) Suggest a reason for benzenediazonium chloride good solubility in polar solvents. [2]
- (b) Aqueous benzenediazonium chloride, $\text{C}_6\text{H}_5\text{N}_2\text{Cl}$, decomposes according to the equation:



The rate of the reaction can be followed by measuring the volume of nitrogen evolved at different times. The following results were obtained.

Time/ min	Volume of N_2/cm^3
0	0
7	54
14	87
21	139
25	172
∞	200

- (i) By plotting a graph of volume of nitrogen gas evolved against time, find the order of reaction with respect to benzenediazonium chloride.
- (ii) In this experiment, the kinetics appears to be zero order with respect to water. Suggest a reason.
- (iii) Calculate the mass of benzenediazonium chloride used in the above reaction, assuming room temperature and pressure conditions. [8]
- (c) A Compound **P** contains C, 35.0%; H, 6.6% and Br, 58.4% by mass. The relative molecular mass of **P** is 136.9. Treatment of **P** with warm aqueous sodium hydroxide gives **Q**, which on oxidation gives **R**. **R** forms an orange precipitate with 2,4- dinitrophenylhydrazine, and both **Q** and **R** on treatment with aqueous alkaline iodine, give a yellow precipitate. When **P** is treated with ethanolic sodium hydroxide it forms three isomeric compounds, **V**, **W** and **X**, each of which, on treatment with HBr , is converted back to **P**.

Deduce, with reasoning, the structures of compounds **P**, **Q**, **R**, **V**, **W**, and **X**.

[10]
[Total:20]